JAMS: INSPECTING A MODULAR AUDIO-TO-MIDI PIPELINE

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ABSTRACT

`jams` is an open-source local music-analysis service with a browser interface for inspecting separated audio, beat grids, functional segments, and per-group MIDI. The demonstration focuses on a practical problem: a cascade can produce plausible stems while making note errors that matter during editing. Users can replay the mixture and inspect transcribed notes in aligned piano rolls; separated audio files are available from the service for external inspection. An accompanying retrospective Slakh2100 case study compares two transcribers before and after a fixed separator. Its reported accompaniment F1 favors YourMT3+ over basic-pitch in both conditions, but does not establish an advantage over direct mixture transcription. We distinguish this preliminary evidence from the prototype's interface capabilities and identify the archived outputs needed for independent recomputation.

1. PURPOSE AND SCOPE

A modular transcription workflow can provide both editable audio groups and MIDI, allowing a musician to choose which representation to use. Source separation followed by transcription is an established approach [1]. The difficult part for an application is making the resulting errors inspectable: interference, missed attacks, extra notes, and merged instruments can affect the symbolic output without being obvious from a single aggregate quality score.

`jams` combines a local HTTP analysis service and a browser-based annotator. [1] The application targets DJ and music-production workflows, but the numerical comparison discussed here uses synthetic instrumental audio. We do not infer performance on real electronic dance music from that dataset. The contribution of the demo is an integrated inspection workflow and a concrete component-comparison question, not a new separation or transcription architecture.

2. SYSTEM AND DEMONSTRATION

The service returns global key and tempo estimates and optionally beat/downbeat times, functional segments, separated audio, and MIDI. The separation path uses an SCNet variant [2] with drums, bass, other, and vocals outputs. The other group combines non-bass pitched instruments; it is not an instrument label. YourMT3+ [3] provides pitched-note transcription, while a separate local model handles drums. The worker architecture keeps incompatible model environments in separate processes. Initial setup downloads some model artifacts; the application should be installed and warmed up before a demonstration.

The demonstration follows an audio example through three operations. First, the user imports a recording and opens its waveform with the returned beat grid and segment boundaries. Second, the user replays the mixture while inspecting the MIDI piano rolls for missing or extra attacks and pitch errors. Separated audio can be examined in an external player; the browser has no stem-solo mixer. Third, the user adjusts segment boundaries or labels and exports per-group MIDI for further work in a sequencer. These operations expose the output for human inspection; the application does not establish transcription correctness simply by displaying it.

The interface also provides a way to examine an evaluation example with known references. A dense accompaniment passage is useful for discussing what an instrument-agnostic score counts: a correctly timed note can be matched even when instrument identity has been discarded. A bass example illustrates why pitch postprocessing must be disclosed. The project's fixed octave shift is an empirical pipeline choice that still requires an audio/MIDI alignment audit, not a universal property of MIDI bass notation.

3. PRELIMINARY COMPONENT COMPARISON

The project ledger records experiments on the 151-track Slakh2100-redux test split [4]. For each track, non-bass pitched reference stems are summed into other and their MIDI notes are merged. The main metric is per-track onset-and-pitch F1, macro-averaged across tracks, with one-to-one matching at 50 ms and 50 cents, ignoring note offsets [5]. Beat quantization is disabled. Thus the measure does not assess durations, instrument assignments, velocities, or editing effort.

Replacing basic-pitch [6] with YourMT3+ raises the reported mean by 0.3591 on reference input and 0.3144 after the same separator (Table 1). The archived reference-input paired 95% bootstrap interval is [0.3471, 0.3713]; no paired end-to-end interval is available. The transcriber ordering persists after separation in this configuration. A

[1] `https://github.com/jhurliman/jams` (distinct from the annotation library `https://github.com/marl/jams`).

Other input	basic-pitch	YourMT3+
Reference group	0.4897	0.8488
SCNet output	0.4733	0.7877

Table 1. Reported mean onset-and-pitch F1 for accompaniment, $n = 151$. These are historical summaries, not newly reproduced measurements.

stronger transcriber on estimated stems exceeding a weaker model on reference stems is not evidence of exceeding an oracle bound.

The experiment does not isolate architecture: YourMT3+ and basic-pitch differ in training data, capacity, decoding, and inference cost. YourMT3+ includes Slakh training data. Component choices were informed by Slakh test results, so these are retrospective development observations. A same-checkpoint direct-mixture YourMT3+ control, with an explicit instrument-to-group mapping, is needed to measure whether separation adds value.

A separate historical separator comparison holds the transcribers fixed and reports higher drum SI-SDR for SCNet than HT-Demucs, but lower drum onset F1. Without paired uncertainty or a controlled mechanism test, this is a diagnostic observation rather than a statistically established reversal. It motivates examining downstream notes in addition to listening to separated audio.

4. EVIDENCE AND NEXT STEPS

The public repository contains the service, scoring code, experiment ledger, and aggregate statistics. The per-recording archives supporting this comparison were unavailable for the present revision and still require release. A committed summary snapshot regenerates the paper's presentation; a separate checker rejects missing archives and mismatched track IDs before recomputing stored-score means and intervals. This does not reproduce inference or establish that stored scores were computed correctly from MIDI.

The next evaluation should freeze the pipeline on a development split, compare direct and separated inputs on paired recordings, and measure note offsets, instrument labels, resource use, and editing effort. A real-audio corpus is needed for the intended production domain. The demo invites feedback on which visible errors matter most to musicians and which comparisons would guide component choice.

5. ACKNOWLEDGMENTS AND AI USE

We thank the authors and maintainers of the cited datasets, models, and evaluation software. Claude (Anthropic) generated the original manuscript and assisted implementation and experiment execution under the author's direction. OpenAI Codex assisted source checking, evaluation-code auditing, and substantive revision. This use extends beyond language editing. The named human author is responsible for the submitted work; neither system is an author. The verification limitations are stated in the scientific content.

6. REFERENCES

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